System Design Specification, Nathaniel Johnson

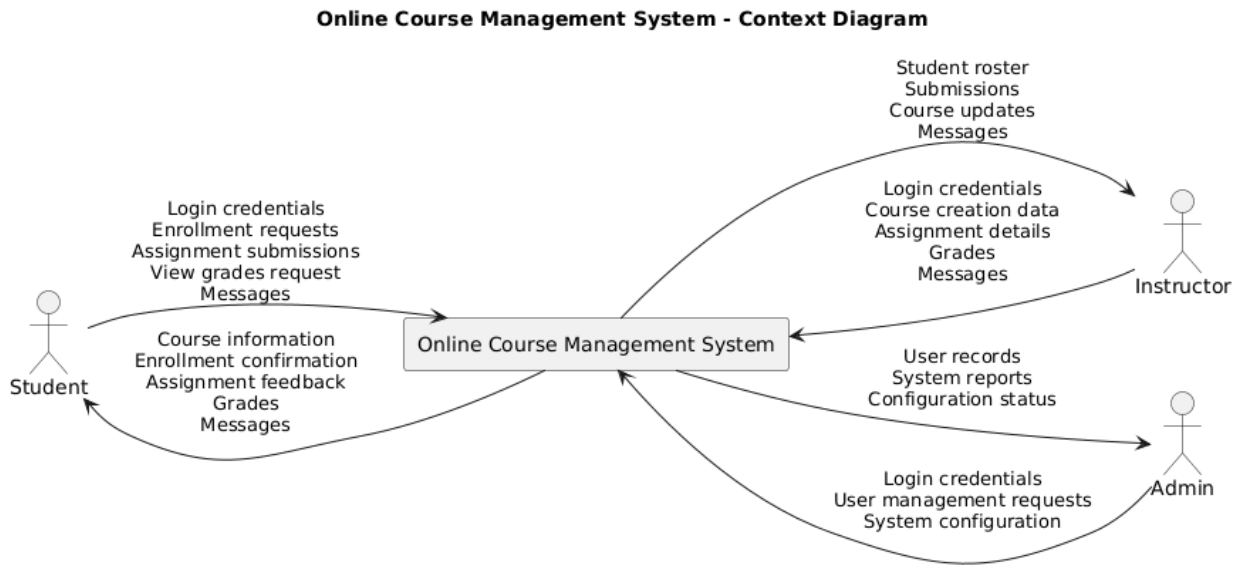
System Overview

The Online Course Management System is a web-based application designed to support academic course management for a university. The system allows instructors to create and manage courses, post assignments, grade student work, and communicate with students. Students can enroll in courses, submit assignments, view grades, and interact with instructors through the system.

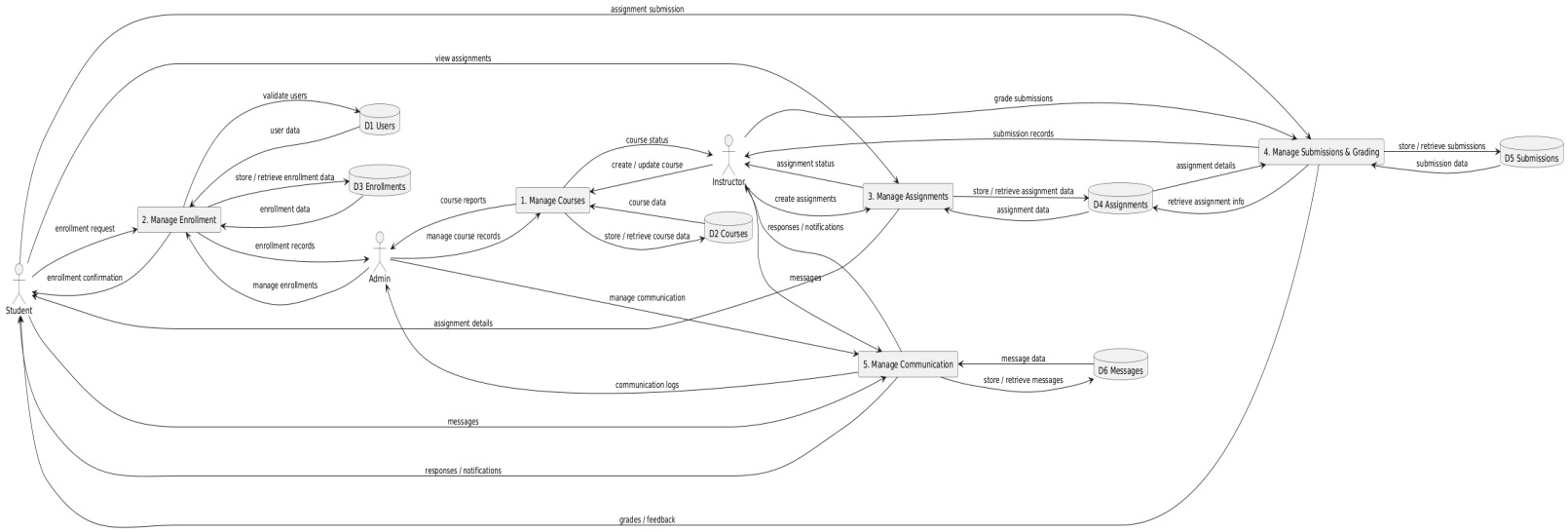
<u>Goals and Objectives</u>

- Provide a centralized platform for course management
- Allow students to enroll and participate in courses
- Enable assignment submission and grading
- Support communication between students and instructors
- Maintain accurate records of academic progress

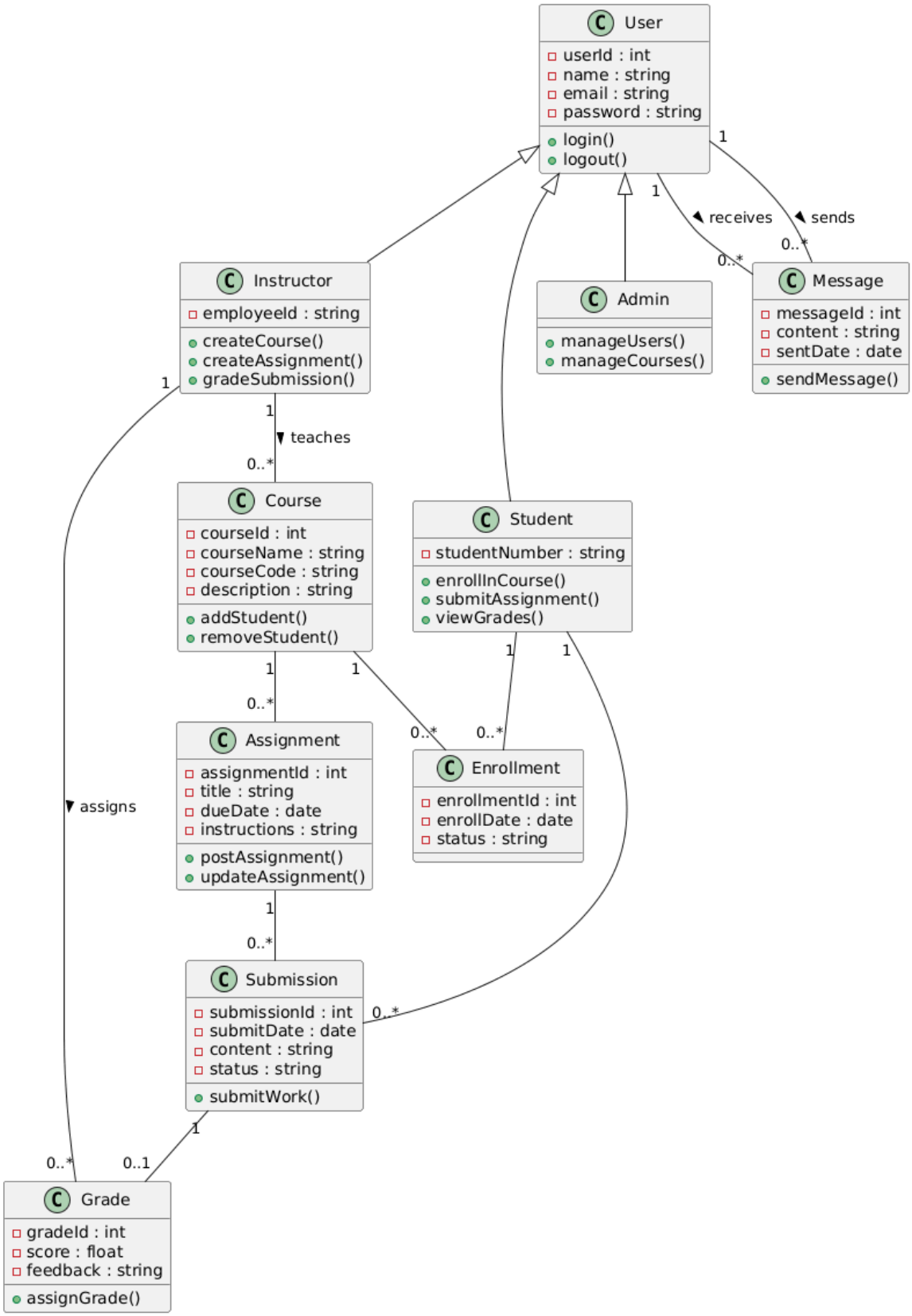
Design Integration



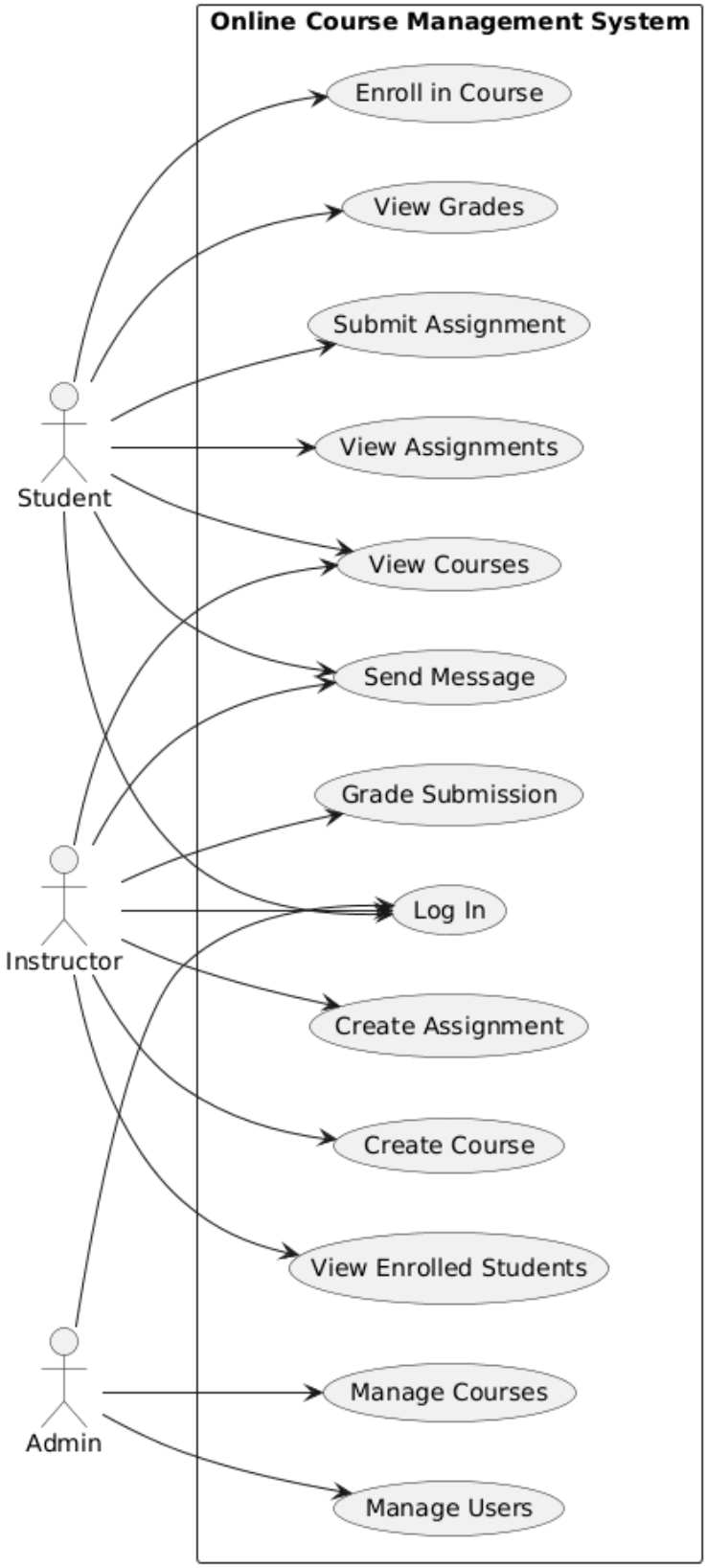
Online Course Management System - Level 1 DFD



Online Course Management System - Class Diagram

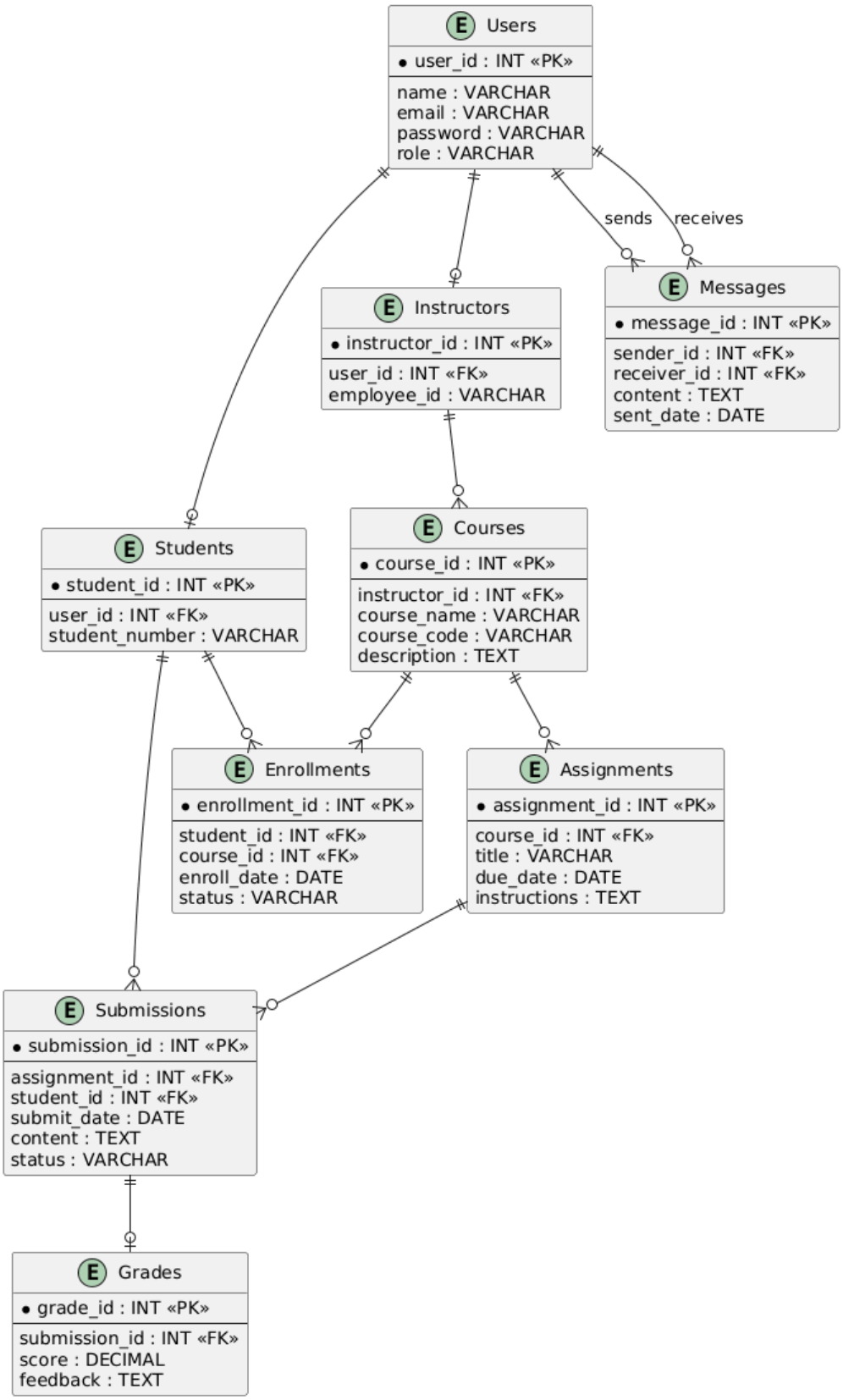


Online Course Management System - Use Case Diagram

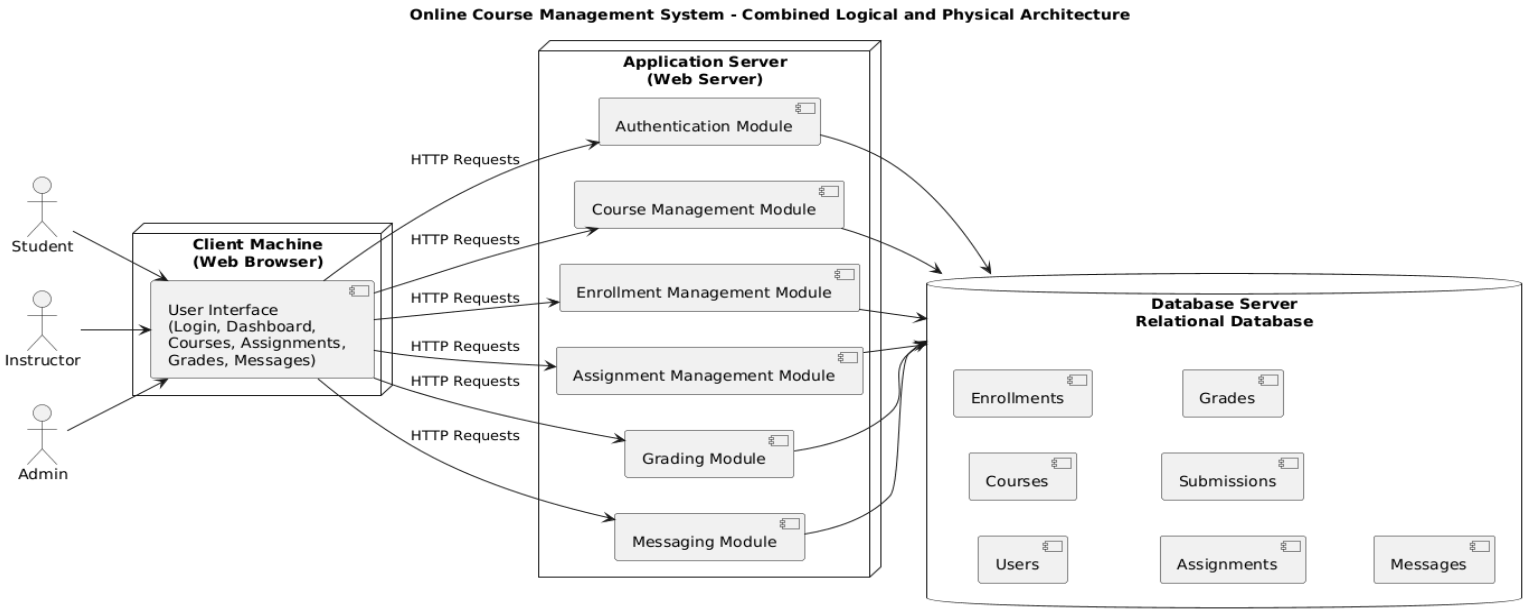


Use Case	**Description**
Log In	Allows a user to access the system using valid credentials
Enroll in Course	Allows a student to register for a course
View Assignments	Allows a student or instructor to see assignment details
Submit Assignment	Allows a student to upload or enter assignment work
Grade Submission	Allows an instructor to review and score student work
Manage Users	Allows an admin to create, update, or remove users

Online Course Management System - ERD



Architecture and UI



<u>UI Mockups</u>

Mockup 1: Student Dashboard

Online Course System | Logout

My Courses

Course: INFO C450

Instructor: Professor Jabin

[View Grades] [View Assignments]

Messages | Notifications

Mockup 2: Assignment Submission Page

Course: INFO C450

Assignment: Project 1

Instructions:

Submit your design document

Upload File: [Choose File]

[Submit Assignment]

Submission Status: Not Submitted

Mockup 3: Instructor Grading Page

Course: INFO C450 – Assignment Submission

Student Name | Submission | Grade | Action

John Doe | View File | [Grade]

Jane Doe | View File | [Grade}

Selected Student : John Doe

Grade: [95]

Navigation Flow and Usability Considerations

The navigation flow of the Online Course Management System is designed to be simple and intuitive for all users. After logging into the system, users are directed to a dashboard based on their role. Students begin at the student dashboard view, where they can select a course. From the course page, they can navigate to view assignments, submit assignments, or view grades. After submitting an assignment, the system displays a confirmation and allows the student to return to the course page or dashboard. Instructors begin at the instructor dashboard view, where they can select a course they teach. From there, they can create assignments, view student submissions, and grade assignments. After grading, instructors can return to the course page or continue grading additional submissions.

The Online Course Management System is designed with usability in mind to ensure that users can interact with the system efficiently and effectively. The system uses a clear and consistent interface layout, with navigation elements placed in predictable locations such as the top or side of each page. Buttons and links are clearly labeled to indicate their function, reducing user confusion.

Security and Risk Considerations

The Online Course Management System is designed with usability in mind to ensure that users can interact with the system efficiently and effectively. The system uses a clear and consistent interface layout, with navigation elements placed in predictable locations such as the top or side of each page. Buttons and links are clearly labeled to indicate their function, reducing user confusion. The system also uses secure database access, where the application communicates with the database through controlled queries. This reduces the risk of unauthorized data access or modification.

The Online Course Management System has taken into consideration many risk factors and mitigation strategies to prevent such risks. The system also uses secure database access, where the application communicates with the database through controlled queries. This reduces the risk of unauthorized data access or modification. Another risk is data loss or corruption, which could occur due to system failures or incorrect updates. This risk is mitigated by maintaining regular database backups and using reliable database management systems.

Design Patterns and Best Practices

The design of the Online Course Management System incorporates several software design patterns and best practices to improve system organization, maintainability, and scalability. The system follows the Model-View-Controller design pattern, which separates the application into three main components: the model, the view, and the controller. The Model represents the data layer of the system, which includes entities such as users, courses, assignments, and submissions stored in the database. The View represents the user interface, which is implemented using web pages that display information to users. The Controller handles the application logic by processing user requests, interacting with the database, and returning the appropriate responses to the user interface.

The system also uses the Client-Server design pattern, where the client communicates with a server to request services and retrieve data. The client is responsible for displaying the user interface and collecting user input, while the server processes requests, executes business logic, and interacts with the database. The server then sends responses back to the client for display.

In addition to design patterns, the system follows several best practices in software design. The system uses modular design, where different functionalities such as course management, enrollment, assignments, and grading are separated into distinct components. This improves code organization and makes the system easier to maintain. The system also follows separation of concerns, ensuring that the user interface, business logic, and data storage are handled independently. This reduces complexity and improves system reliability.